Faithful group actions on a bipartite graph

Private recovery of original L059

13 October 2026

Abstract

A fixed connected bipartite graph carries a finite collection of faithful rank two coset geometries. Their classification is exactly a classification of edge-transitive subgroups of the group preserving the two parts, up to conjugacy in that group. The correspondence retains the acting group and the ordered stabilizers. We give the correspondence and its action invariants, and prove an exhaustive enumeration by descending through maximal subgroups. We apply it to the graph families considered by De Saedeleer, Leemans, Mixer and Pisanski. The retained calculations classify 4254 markings on 1950 inputs, including the subdivisions added in the published edition. Every entry of the four printed tables is recovered, with an explicit correction to the quartic input threshold and an explicit choice of ordered parts. The historical quartic identification uses attributed classification premises; it is not a classification of all semisymmetric quartic graphs.

1 The objects and their kernels

The source of this application is [1]. Its structural results belong to the classical relation between flag-transitive incidence geometries and coset actions, which the source attributes to Tits and subsequent work. We retain that attribution. The proofs of the kernel, marking correspondence, duality and local invariant statements rework the project's existing phase177 argument; they do not count as a second recovery. The maximal-subgroup descent and the subdivision extension lemma give the present organization of the finite computation. Both use elementary group-action facts; we make no claim that those underlying facts are new.

All groups and graphs in the application are finite. A marked coset geometry is an ordered triple $(G; P, Q)$ of a group and two subgroups. Its vertices are the left cosets of P and of Q, kept as distinct types; two vertices of different types are adjacent when their cosets intersect. A marked isomorphism is a group isomorphism that carries the first displayed subgroup to the first, and the second to the second. The action of G is left multiplication on the cosets.

We first make explicit the elementary graph facts used in the construction. Suppose a group G is edge-transitive on a connected graph with at least one edge. Fix an edge $\{u, v\}$. Every vertex is an endpoint of an edge, and hence belongs to the orbit of u or the orbit of v. There are therefore at most two vertex orbits. If there are two, each edge joins the two orbits and the graph is bipartite. If there is one, the group is vertex-transitive; it may or may not be arc-transitive. In the latter case the arcs form two orbits exchanged by reversal. Each has one arc per undirected edge. If the common valency is d and the number of vertices is n, each orbit contains $nd/2$ arcs. Vertex-transitivity makes the number of arcs in that orbit with any prescribed initial vertex equal to $d/2$. Thus d is even. This proves the source's trichotomy and even-valency statement. The stronger last sentence of its Proposition 1 is false for a specified group action: the cyclic group of order three on a triangle is vertex- and edge-transitive but has two orbits on arcs.

A connected bipartite graph has exactly two choices of ordered bipartition. Starting at a vertex, the parity of path length fixes the part of every vertex, so every graph automorphism

either preserves both parts or exchanges them. Its colour-preserving subgroup is consequently the whole group or has index two. These are the connected assertions of the source's Propositions 4–6. They also show the point- and line-transitivity assertion of Proposition 7 in the present connected setting. Indeed, if the full group is edge-transitive, two vertices in the same part can be mapped to one another by a full automorphism: in the two-orbit case these parts are precisely the orbits; in the one-orbit case use vertex-transitivity. Such an automorphism preserves the parts, so the colour-preserving subgroup is transitive on each part.

The disconnected assertion in Proposition 5 needs an additional restriction. For example take three disjoint copies of $K_{1,2}$, colour the centres of two black and the third white. Each component has no colour-exchanging automorphism, since the centre has a different degree from its leaves. An automorphism can nevertheless exchange a black-centred component with the white-centred component while fixing the third component. It then neither preserves both global parts nor exchanges them. Our use of the proposition is confined to connected graphs.

Proposition 1. *Set $B = P \cap Q$ and $N = \mathrm{core}_G(B)$. The action on the union of the two vertex types and the action on the edges both have kernel N. The marking $(G/N; P/N, Q/N)$ has the same incidence graph and acts faithfully on it. The graph has $[G : \langle P, Q\rangle]$ connected components. In particular, it is connected precisely when $\langle P, Q\rangle = G$.*

Proof. An adjacent pair has the form (gP, gQ): choose g in the nonempty intersection of its cosets. Two choices g, h give the same pair precisely when $h^{-1}g \in P \cap Q$. Thus the edge set is the G-set G/B, and its kernel is $\mathrm{core}_G(B)$. The kernels on the two types are $\mathrm{core}_G(P)$ and $\mathrm{core}_G(Q)$. Their intersection is $\mathrm{core}_G(P \cap Q)$, since conjugation preserves intersection. Because N lies in both stabilizers, the maps $gP \mapsto (gN)(P/N)$ and $gQ \mapsto (gN)(Q/N)$ are bijections and preserve nonempty intersections. The quotient kernel is trivial.

Write $L = \langle P, Q\rangle$. Moving from one edge to another through their common vertex changes a representative by right multiplication by an element of P or Q. Consequently every edge in the component of (P, Q) is represented by an element of L. Conversely, a word in elements of P and Q gives a sequence of edges sharing successive vertices, so every such representative is in that component. Translating by g gives the component represented by gL. This proves the component formula. □

Here faithfulness is faithfulness on both types together. Neither individual vertex action is required to be faithful. Proposition 1 gives the source's Theorem 1, Corollary 1, and the connectivity and flag-stabilizer formulas used in its construction.

2 The classification over a fixed graph

Let X be a connected bipartite graph with ordered parts X_0, X_1, each containing at least two vertices. Put $A = \mathrm{Aut}^+(X)$, the group of graph automorphisms preserving each part, and fix an edge $e = \{b, w\}$ with $b \in X_0$, $w \in X_1$. We consider subgroups of A that act transitively on the edges.

Theorem 2. *The assignment*

$$H \longmapsto (H; H_b, H_w)$$

induces a bijection from the A-conjugacy classes of edge-transitive subgroups of A to the marked-isomorphism classes of faithful coset geometries whose ordered incidence graph is isomorphic to X.

Proof. An edge-transitive subgroup H is transitive on each vertex type, because each vertex belongs to an edge. The maps $hH_b \mapsto h(b)$ and $hH_w \mapsto h(w)$ are bijections from the two coset spaces to the vertex parts. They send an intersecting pair to an edge. Conversely, every edge

is an image of e under some $h \in H$, so every adjacent pair comes from intersecting cosets. This is a graph isomorphism. The action is faithful because H is an actual subgroup of the graph automorphism group.

Conversely, use a type-preserving graph isomorphism to transport the faithful action of a marking to X. Its image is a subgroup $H \leq A$. The coset action is transitive on adjacent pairs, by the first paragraph of Proposition 1, hence H is edge-transitive. Choosing the base edge gives exactly its two vertex stabilizers.

It remains to verify the equivalence relation. Suppose $H' = aHa^{-1}$ with $a \in A$. Edge transitivity supplies $h' \in H'$ carrying $a(e)$ to e. As both permutations preserve types, $h'a$ fixes b and w separately. Conjugation by $h'a$ is therefore a marked isomorphism.

For the converse, let $\phi : H \to H'$ carry H_b to H'_b and H_w to H'_w. Define

$$f(h(b)) = \phi(h)(b), \qquad f(h(w)) = \phi(h)(w).$$

The stabilizer conditions make both definitions well defined and bijective. The coset description just proved makes their union an incidence-preserving bijection, with incidence-preserving inverse. Thus $f \in A$. Its definition implies $fhf^{-1} = \phi(h)$ on every vertex, so $fHf^{-1} = H'$. This proves both injectivity and surjectivity. □

The proof shows that conjugacy by the pointwise edge stabilizer A_e gives the same equivalence classes on this collection of subgroups. The faithful marking with acting group A is the source's stable marking. A faithful marking is stable exactly when $H = A$: the reverse construction from its graph produces A, and a proper subgroup of a finite group cannot be isomorphic to it.

Without the faithful requirement there are infinitely many marked realizations of the same graph. From any realization form $(G \times C_m; P \times C_m, Q \times C_m)$ for $m \geq 1$. The common factor disappears from the coset graph, whereas the group orders distinguish the markings. This is the source's nonuniqueness result.

3 Equivalence and duality

An abstract group isomorphism and an isomorphism of the underlying incidence graphs do not in general determine a marked isomorphism. The sentence following Proposition 13 of [1] needs this qualification. For example, put

$$G = C_4 \times C_2^2, \quad P = C_4 \times 1, \quad Q = 1 \times C_2^2.$$

Both $(G; P, Q)$ and $(G; Q, P)$ have faithful incidence graph $K_{4,4}$: the factors intersect trivially and every coset of one factor meets every coset of the other. No group automorphism exchanges P and Q, since only the former has an element of order four. This is the counterexample retained from phase177; its already existing Lean file checks the stated factor obstruction, intersection and incidence facts, and is not a formalization of the census algorithm.

Marked isomorphism is also weaker than simultaneous inner conjugacy. In C_2^2, choose independent a, b. An invertible linear map sends $(\langle a\rangle, \langle b\rangle)$ to $(\langle a+b\rangle, \langle b\rangle)$, while every inner automorphism is trivial. This proves the existence assertion of the source's Proposition 14. Its Mathieu-group example has the same mechanism: an outer automorphism exchanges two classes of subgroups of order 432 and preserves the relevant class of subgroups of order 192. Our explicit finite witness specifies the automorphism and the conjugating element; an arbitrary conjugating element is not needed for the assertion.

More explicitly, the retained permutation certificate in degree twelve has $G = M_{12}$ of order 95040, subgroups P, Q of orders $432, 192$, and an automorphism ϕ with $\phi(P) = P'$,

$\phi(Q) = Q' = Q^h$. The orbit lengths of P and P' on the twelve points are respectively (12) and $(3, 9)$. They therefore cannot be conjugate even in S_{12}, whereas ϕ is a marked isomorphism. Both pairs generate G, and their intersection has trivial core. The certificate also records actual normal subgroups and complements giving

$$P \cong (C_3^2 \rtimes Q_8) \rtimes S_3, \qquad Q \cong Q_8 \rtimes S_4.$$

The second subgroup orders and all intersections are checked on these explicit permutations. Thus the source-specific Mathieu example is retained as well as the elementary existence argument.

Put $F = \mathrm{Aut}(X)$ and allow its elements to exchange the two parts.

Proposition 3. *For an edge-transitive $H \leq A$, the type-exchanging elements of $D = \mathrm{N}_F(H)_e$, where e is fixed as a set, correspond to the automorphisms of H interchanging H_b and H_w. This correspondence preserves orders. In particular the least order of a marked duality is the least permutation order among these elements of D.*

Proof. Conjugation gives one direction. Conversely, for an automorphism ϕ exchanging the stabilizers, use $h(b) \mapsto \phi(h)(w)$ and $h(w) \mapsto \phi(h)(b)$. As in Theorem 2, these formulas give an automorphism of X normalizing H and fixing e as a set.

The conjugation homomorphism on D is injective. A type-preserving element in its kernel fixes both endpoints of e and commutes with H, so it fixes every edge translate, hence every vertex. If a type-exchanging element were in its kernel, it would imply $H_b = H_w$. Connectedness gives $H = \langle H_b, H_w \rangle$, so both vertex parts would be singletons, contrary to our assumptions. Injectivity proves the assertion about orders. □

In the excluded one-edge graph, a permutation of the two types must be retained as additional data: conjugation on the trivial acting group does not detect it. Every cubic or quartic input in the source lies in the nondegenerate case above.

4 Subdivisions and reuse of the group actions

Let X now be a connected cubic graph, and write $S(X)$ for its subdivision: each original edge $\{u, v\}$ is replaced by the path u, e, v through a new vertex e. Order the two parts by putting original vertices first and edge vertices second. Their degrees are three and two respectively.

Proposition 4. *Restriction to the original vertices identifies $\mathrm{Aut}(S(X)) = \mathrm{Aut}^+(S(X))$ with $F = \mathrm{Aut}(X)$. Under this identification, edge transitivity on $S(X)$ is arc transitivity on X. For an arc-transitive subgroup $H \leq F$ and an original edge $e = \{u, v\}$, the corresponding marking has stabilizers*

$$P = H_u, \qquad Q = H_{\{u,v\}}, \qquad B = H_u \cap H_v.$$

Consequently its invariant columns can be computed on X itself.

Proof. Degrees force every automorphism of $S(X)$ to preserve its two parts. Two original vertices are adjacent in X exactly when they have a common edge vertex in $S(X)$. Thus restriction gives an automorphism of X. Its action on each edge vertex is forced by the two endpoints, and every automorphism of X extends in this way. An edge of $S(X)$ is an incident pair $(u, \{u, v\})$, in bijection with the arc (u, v) of X. This bijection is equivariant. The stabilizer formulas follow, since fixing u and the set $\{u, v\}$ is equivalent to fixing u and v separately. □

For the four Max entries, the corresponding coset actions can be computed directly as

$$P \curvearrowright X(u), \quad Q \curvearrowright \{u, v\}, \quad H \curvearrowright V(X), \quad H \curvearrowright E(X).$$

For example, $pB \mapsto p(v)$ identifies the first left coset action, and $hQ \mapsto h(e)$ identifies the last. The other two maps use u. These are equivariant bijections because the point stabilizers are exactly B, B, P, Q, respectively. They preserve primitivity and double transitivity, including for actions with a kernel. The separate component *Computing the subdivision columns on the original graph* gives the complete argument in the right-action convention of GAP and records the exact change to the finite implementation. No classified subgroup changes.

For a bipartite X this computation can reuse its entire earlier classification of edge-transitive subgroups. Put $A = \mathrm{Aut}^+(X)$. Since F is arc-transitive in the symmetric cubic census, $[F : A] = 2$.

Lemma 5. *Every arc-transitive subgroup of F has the form*

$$H = \langle K, x \rangle, \quad K \leq A \text{ edge-transitive}, \quad x \in N_F(K) \setminus A, \quad x^2 \in K.$$

Conversely every such pair gives an arc-transitive H, and $H \cap A = K$. To list the F-conjugacy classes of H, it is enough to use one K from each previously listed A-conjugacy class, test a transversal of K in $N_F(K)$, and deduplicate the resulting subgroups under F-conjugacy.

Proof. If H is arc-transitive, it contains an element exchanging the two parts. Its subgroup $K = H \cap A$ is normal of index two. Given two edges, direct their arcs from the first part to the second. An element of H carrying one directed arc to the other preserves the parts, so lies in K. Thus K is edge-transitive. Any $x \in H \setminus A$ normalizes K, has square in K, and generates H together with K.

Conversely, the hypotheses make $K \cup Kx$ a subgroup, whose intersection with A is K. The two orientations of edges are separate transitive K-sets, and x interchanges them. Their union is therefore transitive under $\langle K, x \rangle$. Replacing x by another element of its K-coset does not change this subgroup or the square condition. Finally, conjugating any K into its retained A-representative also conjugates the associated H and x. The transversal construction therefore misses no F-conjugacy class. Deduplication gives precisely the equivalence relation in Theorem 2, applied to $S(X)$. □

5 The invariant columns

For a marking obtained from H, write $P = H_b$, $Q = H_w$ and $B = P \cap Q$. The four inclusions in the source's Max column are

$$B < P, \quad B < Q, \quad P < H, \quad Q < H.$$

For any inclusion $K < L$, act on L/K. The entry is $2T$ if the action is doubly transitive, M if primitive but not doubly transitive, and X otherwise. This makes sense even when the action has a kernel.

For completeness, primitivity is equivalent to maximality of K. If $K < J < L$, then J/K is a nontrivial block through the base coset. Conversely, let Δ be such a block and let J be its setwise stabilizer. Whenever $gK \in \Delta$, the blocks $g\Delta$ and Δ meet at gK, hence agree. Thus $g \in J$, and $\Delta = J/K$. The possible blocks are exactly the intermediate subgroups. Double transitivity is equivalently the condition that K has two orbits on L/K.

The four Norm entries test whether H, P, Q, B respectively are normal in A. The MaxOrd entry is the greatest element order in H, which can be computed using one representative from each element-conjugacy class. Proposition 3 gives the Dual entry. Using arbitrary dualities of the unmarked graph would lose the required normalizer condition.

Corollary 6. *For a cubic graph the first two Max entries cannot be X. For a quartic graph they cannot be M.*

Proof. In the cubic case both inclusions have prime index three, so no intermediate subgroup exists. In the quartic case the faithful local image is a transitive subgroup of S_4. Its order is one of $4, 8, 12, 24$. An order-four action is regular and has a block system from any subgroup of order two. An order-eight subgroup is a Sylow two-subgroup and preserves a partition into two pairs. The remaining possibilities are A_4 and S_4, both doubly transitive. Thus there is no primitive local action that is not doubly transitive. □

6 Exhaustive descent through maximal subgroups

The classification need not enumerate every subgroup of A before testing edge transitivity. The following elementary observation is the relevant reusable part of the present computation.

Theorem 7. *Suppose A is edge-transitive. Start with the class of A. For each retained subgroup L, take representatives of all its conjugacy classes of maximal subgroups. Retain those that are edge-transitive, except for classes already retained up to conjugacy in A. Continue until every retained subgroup has been processed. The output is exactly one representative of every A-conjugacy class of edge-transitive subgroups.*

Proof. All retained subgroups satisfy the required property, by construction. Every proper step strictly decreases group order, and the finite group A has only finitely many subgroups, so the process terminates.

Let $K \leq A$ be edge-transitive. Choose a chain

$$K = L_s < L_{s-1} < \cdots < L_0 = A$$

in which each inclusion is maximal. Every L_i is edge-transitive, because it contains K. Inductively suppose the class of L_i has a processed representative. Conjugate the next subgroup along with L_i to place it in that representative. A further conjugation inside the representative puts it among the chosen maximal-subgroup representatives. It is therefore retained, or its A-class was already retained. In either case that class is eventually processed. This proves by induction that the class of K occurs. The conjugacy test prevents duplicate output classes. □

If A itself is not edge-transitive, no subgroup can be edge-transitive, and the required list is empty. Otherwise an inexpensive necessary condition is $|E(X)| \mid |L|$; it permits discarding a maximal subgroup before computing its edge orbits. Such a discard is exact by the orbit-stabilizer formula. Failure of edge transitivity is also inherited by every subgroup, which is why no descendant of a discarded subgroup is needed.

Each retained node in our finite receipt records its actual permutation generators and all maximal-subgroup representatives. Each outgoing step records either the failed order test, the failed edge-orbit test, or the index of its retained A-class. The last alternative is checked by actual conjugacy in A, not merely by abstract isomorphism or group order. Exhaustiveness of the finite maximal-subgroup calculation and of the full graph automorphism calculation is an explicit dependency on GAP, GRAPE and nauty. The direct nauty path also verifies every reported generator against the edge set and checks the generated group order in GAP; its completeness still uses nauty's algorithm. This is a computational certificate with a written completeness proof, not a claim of full Lean verification.

7 Finite interfaces and historical scope

The existing five-graph phase177 census has 69 markings: five over $K_{3,3}$, 58 over $K_{4,4}$, and two each over the cube, the Heawood graph, and the subdivision of the Petersen graph. It retains the source's corresponding exceptional examples and the nonstable $A_5 < S_5$ realization. Its witness files are reused rather than counted as new results.

The current primary catalogue gives a further explicit graph on 110 vertices, named $C4[110, 7]$. Its full group is of order 1320 and preserves the two parts. Exhaustive subgroup enumeration gives exactly two markings, with Max entries $2TXXM$ and $2TXMM$. These are the source's two named exceptional codes. The finite receipt records every edge and both stabilizer pairs, so the displayed dihedral notation can be checked through actual group orders instead of an unstated notation convention.

The acting groups and stabilizers are verified by actual group isomorphisms. Writing D_{2m} for the dihedral group of order $2m$, the two markings on 110 vertices are

$$(\mathrm{PGL}(2, 11); S_4, D_{24}), \qquad (\mathrm{PSL}(2, 11); A_4, D_{12}).$$

Their Borel subgroups have orders six and three. The source prints D_6 and D_3; our statement uses the verified orders explicitly. In particular both vertex parts have 55 elements. The remaining unique quartic Max code, $XX2T2T$, occurs on ten vertices with $(C_5 \rtimes C_4; C_4, C_4)$ and trivial Borel subgroup.

The corresponding named cubic cases are checked in the retained five-graph data: the two dual $S_3 \times C_3$ markings over $K_{3,3}$, $(A_4; C_3, C_3)$ over the cube, and $(C_7 \rtimes C_3; C_3, C_3)$ over the Heawood graph. The unique $YYYY$ markings have elementary abelian acting groups C_3^2 and C_2^4 over $K_{3,3}$ and $K_{4,4}$. The same certificates check the unique cubic $YYNN$, $YNYN$ and $NNNY$ cases against the ordered stabilizers. This verifies the group descriptions separately from the aggregate table counts.

Historical input is a separate obligation. The archived 2011 Mini-Census homepage specifies at most 150 vertices. The retained 2008 and 2014 tables both yield 418 bipartite inputs with vertex stabilizer smaller than 100. Thirty are half-arc-transitive and cannot contribute a marking: if F is vertex- and edge-transitive but not arc-transitive, an edge stabilizer fixes its two endpoints; it lies in the index-two group A. Hence the A-orbit of the edge has half the size of its F-orbit. No subgroup of A can be edge-transitive.

The source also explicitly counts $K_{4,4}$, whose full vertex stabilizer has order 144. The literal cutoff therefore does not include an example explicitly present in its table. The 388 graphs with actual vertex stabilizer below 100, together with $K_{4,4}$, yield 1353 markings. Eight additional graphs with vertex stabilizer 128 or 144 yield exactly 290 more. Thus the reproduced table uses 397 graphs and 1643 markings, with the explicit scope

$$|X| \le 150, \qquad |\operatorname{Aut}(X)_v| < 200, \qquad X \text{ bipartite and edge-transitive under } \operatorname{Aut}^+(X).$$

Since the two parts of a regular bipartite graph have the same size, this stabilizer threshold is equivalently $|\operatorname{Aut}^+(X)|/|X| < 100$. It is not the actual vertex-stabilizer threshold stated in the paper.

Under this explicit scope all eleven Norm counts equal the printed counts. The twelve Max counts also agree after reversing the two ordered parts of current graph $C4[128, 55]$, named $SS[128, 16]$. It is semisymmetric: its two color orders need not define isomorphic ordered graphs. This reversal transfers three entries from $X2TXX$ to $2TXXX$ and changes no Norm count. The stored witnesses retain the original coloring, and the aggregation records this reversal explicitly. These calculations establish a reproducible numerical scope; they do not establish why the original table used it.

Historical identification uses the separate, reviewed component *Identifying the historical quartic input set*. Wilson and Potočnik state that their expanded census contains all previously

known edge-transitive tetravalent graphs up to 512 vertices [3, §1]. Acquiring that coverage assertion and the historical census's distinctness and invariant assertions, each historical set in the corrected scope is contained in the current set. The retained 2008 and 2014 tables each give 397 classes, while direct checks of the current edge data give 397 pairwise nonisomorphic classes in that scope. Inclusion and equal finite cardinality therefore identify the sets. This argument does not identify every old tag, order the two parts, or claim completeness among all possible semisymmetric quartic graphs. The 28 explicit isomorphisms reconstructed from the archived BC, MSY and GPS2 definitions remain supplementary witnesses, not substitutes for the stated coverage premise.

The cubic scope is distinguished from the date of the cited classification. The primary Foster site documents extensions from 768 to 2048 vertices and then to 10000. We acquire the completeness of its cubic symmetric and semisymmetric lists from the cited classification, rather than attempting another graph enumeration. The retained primary LCF records reconstruct all 796 symmetric inputs through 2048 vertices and all 43 semisymmetric inputs through 768 vertices. The two non-Hamiltonian symmetric graphs, the Petersen and Coxeter graphs, have explicit separate constructions. Direct checks verify their simplicity, connectedness, valency and pairwise nonisomorphism. Of the 796 symmetric graphs, 714 are bipartite. For 261 earlier bipartite inputs, explicit vertex permutations identify the retained graph and its current LCF realization, so their completed subgroup classifications are reused.

Every required graph now has a completed finite receipt. The 714 bipartite symmetric graphs give 1082 faithful markings; all ten Max and all six Norm entries agree with the printed cubic tables. The 43 semisymmetric cubic graphs give 102 markings. The final 2014 source explicitly includes subdivisions of symmetric cubic graphs, an obligation absent from its arXiv v2 text. All 796 such inputs have been classified and give 1427 markings. Proposition 4 and Lemma 5 compute these on the original vertex sets. Every entry retains the acting permutation group, its two stabilizers, their intersection, and the Max, Norm, Dual and MaxOrd columns. Two initial nonbipartite calculations reached their time limits and made no mathematical conclusion. Their separate successful retries use the direct action calculation; both initial receipts and the changed engines remain in the record.

Input family	Graphs	Faithful markings
Bipartite symmetric cubic, $n \leq 2048$	714	1082
Semisymmetric cubic, $n \leq 768$	43	102
Subdivisions of symmetric cubic, original $n \leq 2048$	796	1427
Historical quartic, corrected scope	397	1643
Total	1950	4254

Table 1: Completed classifications. The scopes are stated above; no large-stabilizer or unbounded classification is asserted.

For the named historical quartic exceptions we do not need to identify every old catalogue label. Within each of the identified historical and current sets, the pairs $(n, |\operatorname{Aut}(X)|)$ equal to $(8, 1152)$, $(10, 240)$ and $(110, 1320)$ each occur once. Since these are graph invariants, equality of the sets identifies these three historical entries with the three explicit graphs used above. This argument does not extend an invariant match to a graph isomorphism without the already established inclusion and uniqueness.

The published edition retains the problem about large stabilizers; the earlier arXiv edition also asks about cyclic configurations. A finite list and the classification theorem above settle neither open-ended classification problem.

$B < P$	$B < Q$	$P < H$	$Q < H$	Count
2T	2T	2T	2T	4
2T	2T	M	M	21
2T	2T	X	X	315
2T	M	M	2T	1
2T	M	X	X	32
M	2T	2T	M	1
M	2T	X	X	32
M	M	2T	2T	1
M	M	M	M	88
M	M	X	X	587

H	P	Q	B	Count
N	N	N	Y	1
Y	N	N	N	404
Y	N	N	Y	674
Y	N	Y	N	1
Y	Y	N	N	1
Y	Y	Y	Y	1

Table 2: Cubic symmetric Max and Norm counts. Every displayed count equals the corresponding printed entry of [1]. The cubic table uses the bipartite symmetric graphs through 2048 vertices.

$B < P$	$B < Q$	$P < H$	$Q < H$	Count
2T	2T	2T	2T	10
2T	2T	M	M	5
2T	2T	X	X	38
2T	X	M	M	1
2T	X	X	2T	11
2T	X	X	M	1
2T	X	X	X	19
X	2T	2T	X	11
X	2T	X	X	26
X	X	2T	2T	1
X	X	M	M	17
X	X	X	X	1503

H	P	Q	B	Count
N	N	N	N	365
N	N	N	Y	268
N	N	Y	N	4
N	N	Y	Y	3
N	Y	N	N	4
N	Y	N	Y	3
Y	N	N	N	466
Y	N	N	Y	525
Y	N	Y	N	2
Y	Y	N	N	2
Y	Y	Y	Y	1

Table 3: Quartic Max and Norm counts. Every displayed count equals the corresponding printed entry of [1]. The quartic threshold and ordered-part correction are explicit in the text.

References

[1] J. De Saedeleer, D. Leemans, M. Mixer and T. Pisanski, *Core-free, rank two coset geometries from edge-transitive bipartite graphs*, Mathematica Slovaca 64 (2014), 991–1006. Fixed primary edition: arXiv:1106.5704v2, 29 June 2011.

[2] S. Wilson and P. Potočnik, *A census of edge-transitive tetravalent graphs*, Mini-Census and Full Census. Exact archived editions and the current graph data are recorded in the accompanying source ledger.

[3] S. Wilson and P. Potočnik, *Recipes for Edge-Transitive Tetravalent Graphs*, arXiv:1608.04158v1 (2016).

[4] M. Conder and P. Potočnik, *Edge-transitive cubic graphs: analysis, cataloguing and enumeration*, Journal of Algebra **685** (2025), 703–737; accompanying census.